

NANO

OmniProbe

Nanomanipulator Family

Superior nanomanipulation for SEM and FIB

MAKE

MOVE

MODIFY

MANIPULATE

MEASURE



The Business of Science®

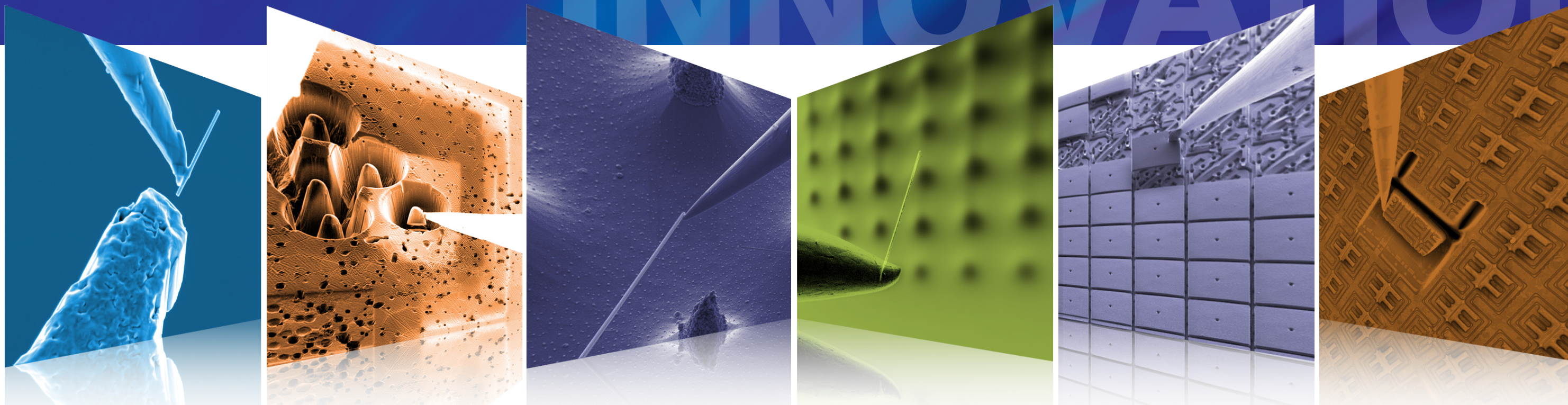


MANIPULATION

OmniProbe Manipulator Family

The most capable and easy to use in situ nanomanipulators...

...optimum system performance



OmniProbe nanomanipulators

Oxford Instruments offers a wide range of manipulator solutions for all budgets and applications: from affordable open-loop servo motor systems to state-of-the-art solutions with closed-loop control and piezo actuation.

Every laboratory can benefit from an OmniProbe in situ nanomanipulator!

In situ solutions for
SEM and FIB applications

Where OmniProbe manipulators excel

OmniProbe nanomanipulators are world renowned for their effectiveness, ease of implementation and use

- Manipulate in the SEM and FIB at the nanoscale
- Neutralise highly charging areas on uncoated insulating specimens
- Measure electrical parameters of nanostructures
- Execute unique FIB lift-out solutions for plan-view and backside milling
- Make correlative TEM and atom probe nanowire samples in the SEM
- Enable high quality results for TEM, EDS, EBSD and other types of analysis
- Perform voltage contrast and electrostatic manipulation using built-in probe tip biasing

Benefits Flexibility

- Port-mounted system maximises microscope performance and capability
- Out of the way when not required, ready to use in seconds
- Maximize interior space savings and avoid conflicts with other accessories
- Retain full functionality of the microscope
- Ideal for all sample sizes and geometries
- Independence from the microscope stage provides extra degrees of freedom for sample manipulation
- Systems available for all budgets

Precision

- Open-loop control enables sub-nm resolution
- Closed-loop control provides repeatable, predictable and intuitive motion with 10nm resolution

Productivity

- Improve sample flow and reduce loadlock or venting cycles
- Perform operations with the fewest steps using unique technology and processes
- Designed for efficient tip replacement, sharpening, and positioning

OmniProbe 100

Manual nanomanipulation

The **OmniProbe 100** is an industry leading, reliable, manual manipulator. Perform in situ lift-out, charge neutralisation, mechanical and electrical testing, and a wide range of other applications.

Features and Benefits

- Speed and flexibility for diverse applications are enabled by a port-mounted interface and open-loop control
- Achieve smooth manipulation over long distances
- Intuitive and easy operation aided by Cartesian movements
- Navigate rapidly by pneumatic insertion with precise motor positioning
- No computer required - operation is performed using a 6-button control box

Flexible,
accurate and
easy to use

OmniProbe 200

Fully automated nanomanipulation



OmniProbe 200

The most popular nanomanipulator for SEM and FIB microscopes. This closed-loop system is completely independent of the microscope's stage positioning to enable extra degrees of freedom and omnidirectionality for sample manipulation.

Features and Benefits

- Enable diverse applications from nanowire manipulation, to TEM sample preparation, to electrical and mechanical measurements
- Smooth, precise nanomanipulation over long distances is enabled by 100 nm closed-loop feedback
- Computer control with closed-loop encoder feedback makes manipulation simple, repeatable and intuitive
- Make true x, y and z axis movements using the stage frame of reference at any stage tilt and port installation
- Unique solutions leverage combined stage and probe (CSP) recipes to achieve fast sample repositioning to specific angles

Omnidirectional travel

MANIPULATION

OmniProbe 400

The ultimate all-in-one nanomanipulator...

...optimised for nanometre accuracy and high speed



OmniProbe 400

A new generation of nanomanipulator, the OmniProbe 400 exploits innovative piezo actuation for best-in-class nanoscale positioning.

Optimised for maximum flexibility and performance, the OmniProbe 400 is the ultimate nanomanipulator for high resolution and high throughput applications.

Unlike any other manipulator, the OmniProbe 400 provides closed-loop motion in 4 degrees of freedom as standard: tip rotation, tip sharpening and sample positioning are now easy for any user.

<1 nm
resolution

4³ performance

Nanoscale manipulation and testing in the SEM, and varied lift-out TEM sample preparations in the FIB can be performed easily while using optimised working distances.

- 4 degrees of freedom (X, Y, Z, R)
- 4 mm or more working distance
- 4 motion strengths: encoder resolution, motor resolution, repeatability, linearity

Concentric rotation

The tip stays in the field of view for rapid tip shaping and sample reorienting without the risk of sample loss or accidental venting

- Position with sub-degree accuracy
- Extend the needle life
- Create fine tips exhibiting ohmic behavior
- Prepare microtips for nanoparticle manipulations that use Van der Waals forces
- Reorient samples for plan-view, side-view and backside preparations

Accuracy

The OmniProbe 400 uses <1nm open-loop and 10 nm closed-loop control to deliver smooth motion and dependable accuracy

- Move diagonally in any direction
- Navigate orthogonally at any sample tilt
- Rotate concentrically
- Always know where the probe tip is, even when outside the field of view
- Move predictably on any port
- Avoid collisions
- Move with high accuracy, linearity and repeatability
- Use fixed and user-programmable positions
- Change tips in situ

Speed

Automation on multiple levels enables results in the fewest steps possible

- Insertion repeatability eliminates tip realignment
- Change tips in 3 minutes without venting
- Concentric rotation enables rapid tip sharpening, tip cleaning, and sample repositioning
- Safely and precisely navigate to stored positions close to the sample and back to park

Integrated coarse and fine motion

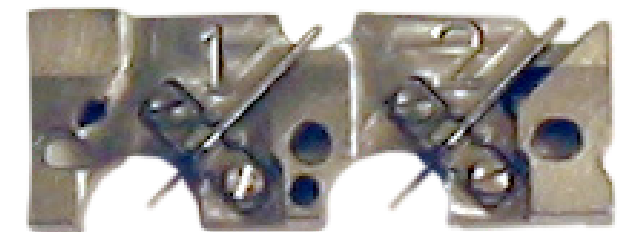
Coarse and fine motion in the same actuator allows smooth transitioning from coarse long moves to fine steps at maximum precision

- Velocity control achieves fast moves over long distances, or slow moves for nanoscale navigation
- Step control provides efficient and precise navigation over a specific distance

Change tips in
<3 minutes!

In situ tip change

- Probe needles can be replaced in under 3 minutes without venting the microscope
- Atmospheric contamination is avoided and the high vacuum preserved for the best performance
- Samples attached to tips can be safely stored for later use, allowing multiple users to maintain their throughput



Linear motion in 3D space

Automation of all three axes enables simple and intuitive navigation, regardless of hardware configuration

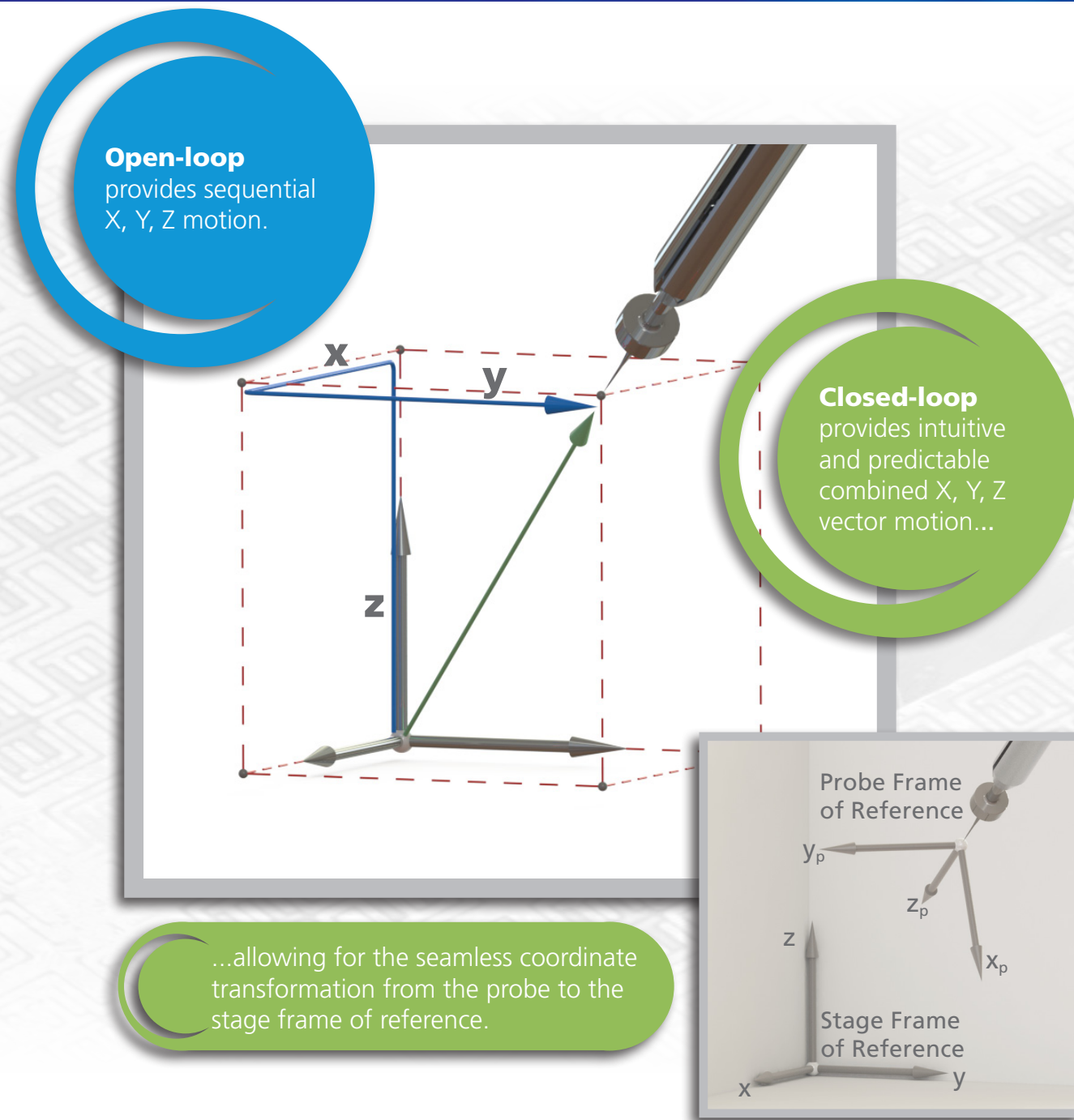
- Move in a straight line in any direction
- Navigate by Cartesian coordinates
- Compound axes control enables motion normal to the stage at any degree of tilt

OmniProbe Manipulator Family

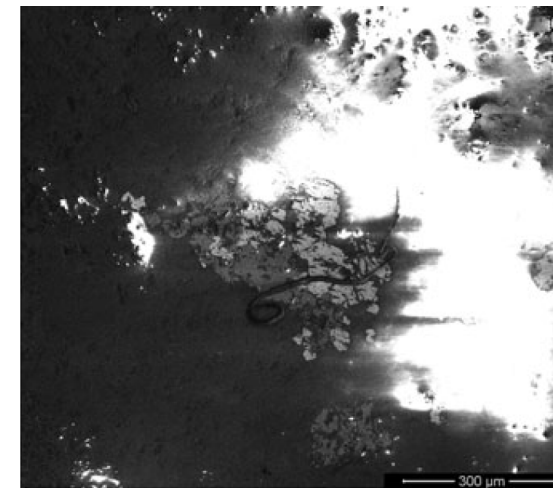
Superior hardware design provides optimum system performance

Improve Imaging

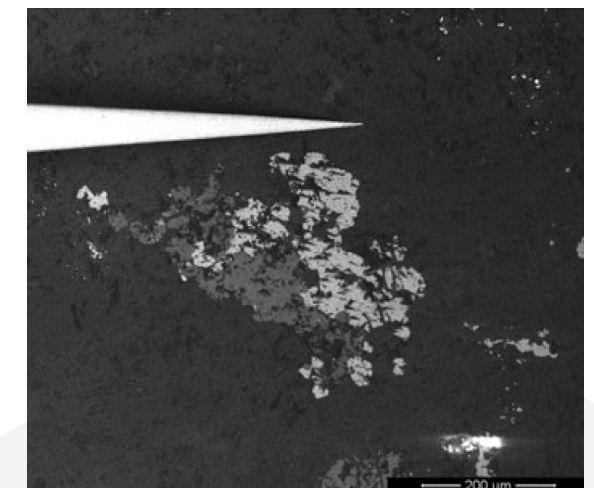
Reduce charging and reveal features



Charge neutralisation

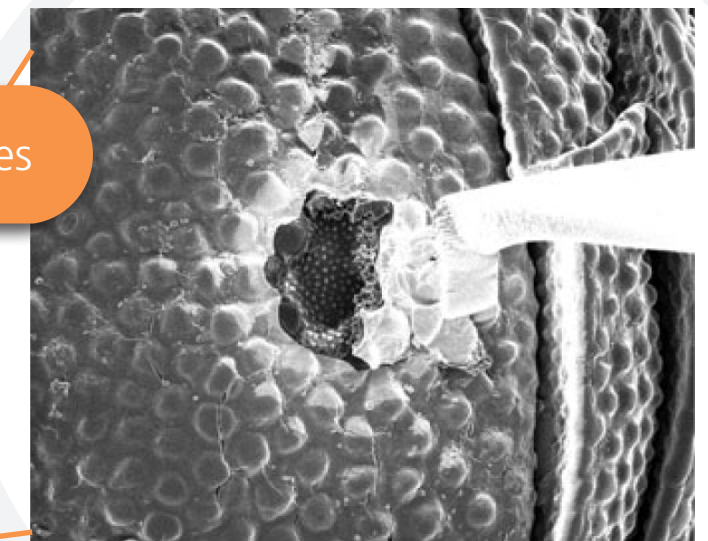
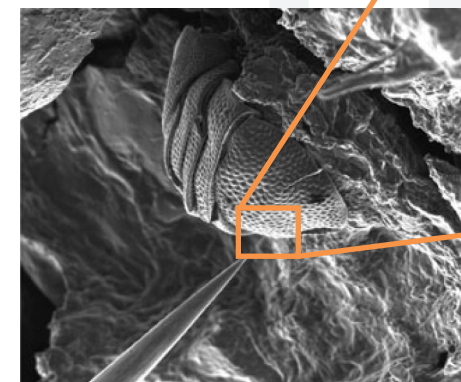


An uncoated sample of rock varnish is difficult to image due to significant charging ...



... contact with a needle neutralises the charge to enable higher magnification and improved quality.

Uncover hidden features

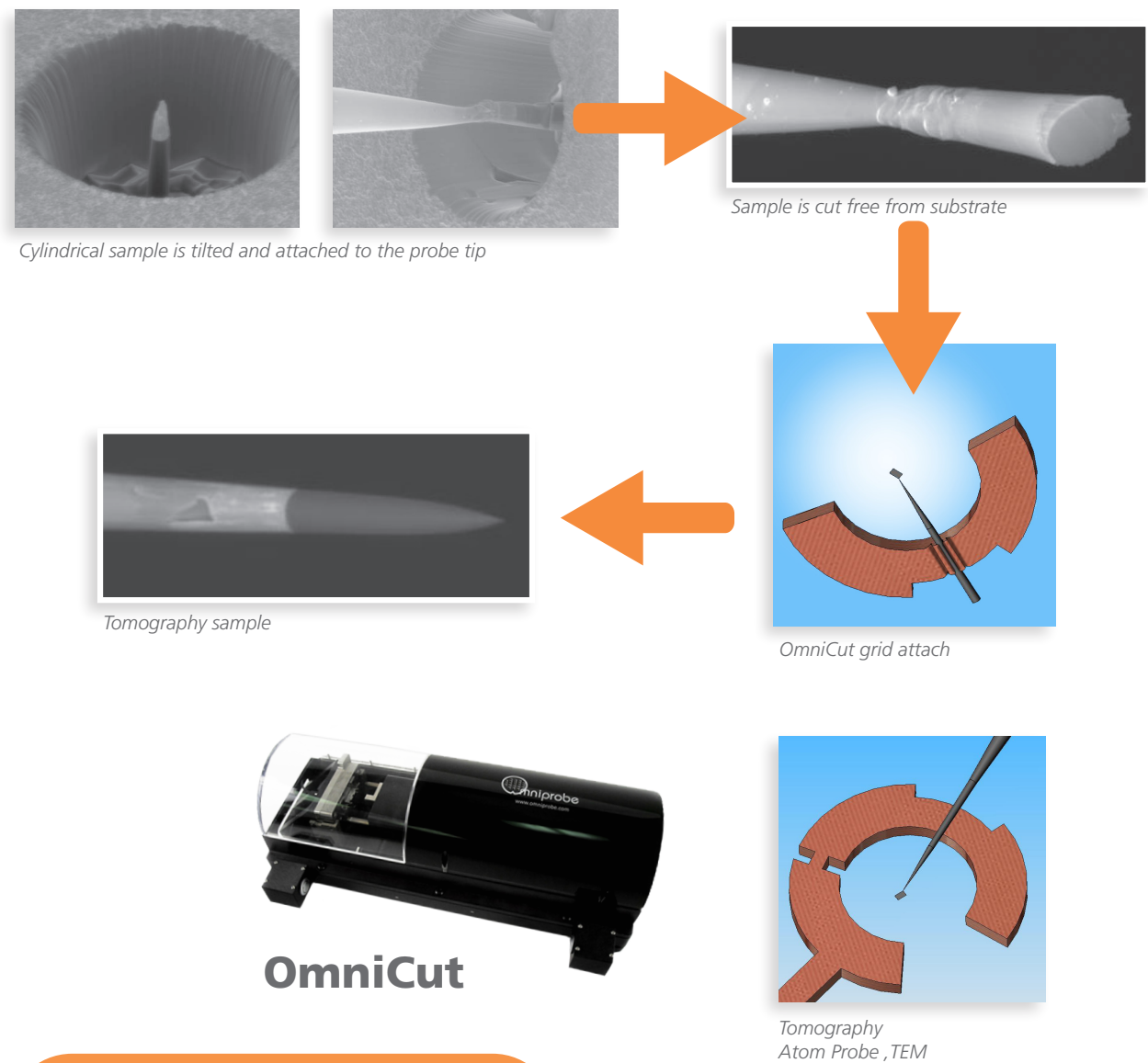


Easily perform microdissection of materials such as biological samples and MEMS devices using precise and accurate positioning to uncover features.

Make Analytical Samples

Extending FIB Capabilities

When combined with the OmniCut tool, the OmniProbe 200 or 400 nanomanipulators easily generate inverted atom probe samples.



Create tomography specimens

	OP100	OP200	OP400		
Features				Benefit	Application
Port-mounted	I	I	I	- Consumes minimal chamber space - Installed once, remains for on-demand use	- TEM preparation of whole wafers - Manipulating with multiple detectors/accessories inside chamber
Linear motion in ANY direction	NA (XYZ only)	I	I	Coordinated motion as well as X, Y, or Z axis moves for efficient operation to and from the sample	TEM preparation, nanowire manipulation, nanostructure assembly
Uninterrupted wide range of travel at high and low velocity	I	I	I	Efficient operation, higher throughput, improved handling and success rates for all manipulation tasks	Manipulation of nano-objects over microns of distance
Closed loop encoder feedback	NA	100nm	10nm	- Optimized accuracy, repeatability, and linearity. - Coordinated motion of multiple axes for navigation by stage frame of reference. Stored locations.	TEM and atom probe sample preparation, including plan-view samples. Manipulation of nanoparticles.
In situ tip change	NA	NA	I	3 minute tip exchanges - Avoid chamber venting	- High resolution imaging (chamber is kept clean) - Preparing sensitive samples (no exposure to air) - Electrical measurements (minimize probe oxidation, quickly change contaminated tips)
Concentric rotation	NA	O	I	Digital, closed loop, 360 degree probe tip rotation keeping the tip in view for tip shaping or sample repositioning	- Plan-view sample manipulation - Sideways thinning - Backside thinning - Electrical testing (ohmic contacts) - Moving nano-objects with Van der Waal's forces
Built-in needle biasing	C	I	I	Float, ground, or apply voltage in the range of -10 volts up to +10 volts	Voltage contrast, charge neutralization, electrostatic manipulation
Electrical feedthroughs*	I	I	I	Connect to various electronics	Electrical measurements with equipment supplied by the user
Electrical test package	NA	O	NA	Measure current or resistance using one probe and microscope's stage electrical connections	- Voltage contrast - EBIC/EBAC - I-V curves
OmniCut accessory	NA	O	O	Perform grid attach outside of the FIB	Prepare TEM-compatible atom probe samples

*Feedthroughs are built into the manipulator and do not require an extra port

I - Included O - Optional NA - Not Applicable C - Compatible

OISERVICE

Global Customer Support

Accredited, experienced, responsive, dedicated

Oxford Instruments recognises that your success requires not just only world-class products, but also world-class service and support. Our global service team is renowned for delivering outstanding service to customers and microscope vendors:

- Hands-on and theory classroom training
- On-site training tailored to your specific needs
- Web-based courses and training videos
- Consultancy and application support
- Multi-layered maintenance and service contracts



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